Software Requirements Specification

For

AI Chatbot For Legal Advice Based On Indian Laws

Version 1.0 (Approved)

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to describe the functional and non-functional requirements of the *AI Chatbot for Legal Advice based on Indian Laws.* This system is designed to assist individuals by providing accessible, context-aware, and reliable legal information tailored to the Indian legal framework. The chatbot will act as an intelligent virtual legal assistant capable of answering queries, guiding users through relevant legal provisions, suggesting procedural steps, and offering references to statutory laws and regulations. This document serves as a reference for developers, legal experts, project stakeholders, and evaluators to ensure a clear understanding of the system's objectives, constraints, and implementation guidelines.

1.2 Document Conventions

This SRS follows the **IEEE 830-1998** standard for software requirement specifications. The conventions adopted are:

- **Section numbering** follows hierarchical format (e.g., 1, 1.1, 1.1.1).
- **Keywords** such as *shall*, *must*, and *should* indicate different levels of requirement priority.
- **Technical terms** and abbreviations (e.g., AI, NLP, LLM, IPC,CrPC,IT Act) are expanded at their first appearance for clarity.
- **Figures and tables** (to be included in later sections) will follow consistent labeling for cross-reference.
- **Legal references** (e.g., Article 21 of the Constitution of India, Section 420 IPC) will be cited using standard legal notation.

1.3 Intended Audience and Reading Suggestions

The intended audience for this document includes:

- **Project Developers**: To understand the detailed system requirements, architecture, and implementation scope.
- **Research Mentors and Faculty**: To validate the accuracy of legal content, ensure compliance with Indian laws, and provide domain-specific insights.
- **End Users (Students/Professionals)**: To anticipate the system's functionalities, usability, and reliability as a legal information resource.

- **Future Researchers**: To extend the system for advanced legal reasoning, multilingual support, and integration with evolving AI technologies.

Readers are encouraged to begin with the *Introduction* for project context, proceed to *Overall Description* for functional understanding, and finally explore *Specific Requirements* for detailed implementation.

1.4 Project Scope

The AI Chatbot for Legal Advice based on Indian Laws is designed as an intelligent virtual assistant that provides accessible, context-aware, and reliable legal information. Its scope includes:

- **Legal Information Assistance:** Offering explanations of relevant laws, acts, and constitutional provisions in simplified language.
- **Guided Legal Queries:** Responding to user questions on common legal topics such as contracts, property, family law, cybercrime, and criminal law with references to Indian legal provisions.
- **Multilingual Support:** Delivering responses in English, Hindi, and other regional languages to ensure accessibility across diverse user groups.
- **Contextual Recommendations:** Providing references to sections of the Indian Penal Code (IPC), Code of Criminal Procedure (CrPC), Constitution of India, and other relevant acts.
- **Scalability**: Capable of expanding to cover additional domains of law, integrating updates in legislation, and supporting a wider user base.

The system does not aim to replace professional legal consultation or representation in courts of law but to complement human decision-making by providing quick, reliable, and understandable legal information.

1.5 References

[1] IEEE Standard for Software Requirements Specification (IEEE 830-1998).

[2] OpenAI Documentation – Large Language Models and API Usage (2025).

[3] LangChain Documentation – Framework for Developing Applications Powered by Language Models (2025).

[4] Whisper API Documentation – Automatic Speech Recognition (2025).

[5] Constitution of India, Government of India.

[6] Natural Language Processing (NLP)-Powered Legal ATMs (LAMs) in India: Possibilities and Challenges," Journal of the Knowledge Economy, vol. 14, no. 2, pp. 123-135, 2023. Available: https://doi.org/10.1007/s13132-023-01450-2

[7] "Artificial Intelligence for Legal Chatbot," Journal of Engineering Sciences and Technology, vol.15, no.9, pp.25-30,2024.

2. Overall Description

2.1 Product Perspective

The *AI Chatbot for Legal Advice based on Indian Laws* is an AI-driven system designed to serve as an intelligent legal information assistant. Unlike conventional chatbots that provide generic responses, this system integrates domain-specific knowledge of Indian legal frameworks, statutes, and procedures to deliver accurate, context-aware legal information.

The chatbot leverages Natural Language Processing (NLP), Large Language Models (LLMs), and multilingual support to handle user queries in English, Hindi, and potentially other regional languages. It is designed to reference the Indian Penal Code (IPC), Code of Criminal Procedure (CrPC), Constitution of India, Information Technology Act, and other relevant legislations when generating responses.

The system will be developed as a modular application with a user-friendly front-end (web and mobile interface) and a back-end powered by APIs (OpenAI, LangChain, Whisper for voice input, and legal knowledge databases). This architecture ensures scalability, extensibility, and ease of integration with future enhancements such as voice-based interactions, expanded legal domains, and connections to government legal resources.

2.2 Product Features

The main features include:

- **Legal Query Handling –** Respond to user questions on diverse legal topics such as contracts, property, family law, cybercrime, and criminal law with references to relevant Indian statutes.
- **Statutory References –** Provide direct citations from the Indian Penal Code (IPC), Code of Criminal Procedure (CrPC), Constitution of India, Information Technology Act, and other applicable laws.
- **Procedural Guidance –** Suggest step-by-step procedures for filing complaints, drafting applications, and understanding legal processes, while clarifying that it does not replace a licensed advocate.
- **Multilingual Support –** Deliver responses in English, Hindi, and other regional languages to ensure wide accessibility.
- **Voice-Based Interaction –** Allow users to interact with the chatbot using speech-to-text and text-to-speech functionality for convenience.
- **Educational Resources –** Offer simplified explanations of legal concepts for students, professionals, and citizens seeking awareness of their rights and responsibilities.

2.3 User Classes and Characteristics

The *AI Chatbot for Legal Advice based on Indian Laws* is expected to serve multiple user classes, differentiated by their intended usage, technical expertise, and priority:

- **Students** – – Law students or other academic learners seeking to understand Indian legal provisions, statutes, and procedures.
- **Professionals** – Individuals requiring quick legal guidance or clarification on contracts, compliance, or workplace-related legal matters.
- **Researchers/Academicians** – For legal research, comparative analysis, and educational purposes..
- **General Citizens** – For understanding rights, filing complaints, or accessing simplified legal information without formal legal consultation.

Users are expected to have **basic digital literacy** and the ability to read and comprehend English or Hindi. The interface will be intuitive, user-friendly, and accessible, ensuring smooth interaction regardless of prior legal knowledge.

2.4 Operating Environment

- **Operating System:** Linux, Windows, or macOS for development.
- **Programming Languages:** Python (due to its rich ecosystem for NLP and ML).
- **Database/Storage:** For storing the knowledge base (e.g.,PostgreSQL,MongoDB) and potentially user data.
- **NLP/ML Libraries:** Tensorflow, PyTorch, scikit-learn, NLTK, spaCy, Transformers (Hugging Face).
- **Hardware Requirements:** Minimum 4 GB RAM, standard CPU, and sufficient storage to handle application data.
- **Web Framework(for UI):** Flask, Django, or React.
- **Cloud Platform (for deployment):** AWS, Google Cloud, Azure (depending on scalability needs).

2.5 Design and Implementation Constraints

- **API Dependency:** The system assumes continuous access to OpenAI APIs for natural language processing and Whisper API for voice recognition. Service interruptions or changes in API policies may impact functionality.
- **Legal Content Accuracy:** It is assumed that statutory and regulatory sources referenced (IPC, CrPC, IT Act, Constitution of India, etc.) are accurate and up-to-date at the time of implementation. Regular updates will be required to maintain accuracy.
- **Internet Connectivity:** Stable internet access is assumed for all real-time API calls, cloud storage access, and updates.

- **User Digital Literacy:** Users are assumed to have basic digital literacy to operate a web-based chatbot, though advanced technical skills are not required.
- **Development Timeline:** It is assumed that the system will be implemented within the academic project timeline. Certain features (e.g., integration with government legal databases or advanced multilingual processing) may be phased into future versions.

2.6 User Documentation

The system will be accompanied by:

- **Quick Start Guide** – Step-by-step instructions for installation and first use.
- **User Manual** – Detailed explanation of features with screenshots.
- **FAQ Section** – Solutions to common issues.
- **Inline Help** – Tooltips and in-app guidance for first-time users.
- **Tutorial Videos (Optional):** Short demos of key features (MP4/WebM

2.7 Assumptions and Dependencies

- Users will have **stable internet connectivity** for API-based features.
- Assumes supported hardware, OS, and browser availability.
- The project depends on third-party libraries such as **LangChain, OpenAI APIs, Whisper API, and Streamlit**.
- Future enhancements may integrate **cloud storage, mobile applications, and multi-user support**.
- Legal references (IPC, CrPC, IT Act) are assumed accurate and current.

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3. System Features

3.1 Legal Query Handling

3.1.1 Description and Priority

This feature allows users to submit legal questions in natural language and receive context-aware, statute-backed responses. It provides accurate, simplified explanations of Indian laws.
Priority: High

Component Ratings: Benefit = 9, Penalty = 8, Cost = 6, Risk = 5

3.1.2 Stimulus/Response Sequences

- **Stimulus:** User types or speaks a legal question.
- **Response:** System processes the query using NLP/LLM and returns a relevant legal explanation, references applicable statutes, and optionally provides procedural guidance.

3.1.3 Functional Requirements

- **REQ-1:** The system shall accept user queries via text input.
- **REQ-2:** The system shall accept user queries via voice input and convert speech to text.
- **REQ-3:** The system shall reference relevant Indian statutes (IPC, CrPC, IT Act, Constitution).
- **REQ-4:** The system shall handle ambiguous or incomplete queries by requesting clarification from the user.

3.2 Procedural Guidance

3.2.1 Description and Priority

This feature guides users through step-by-step procedures for filing complaints, drafting applications, or understanding legal processes.
Priority: High

3.2.2 Stimulus/Response Sequences

- **Stimulus:** User requests guidance on a legal process.

- **Response:** System provides stepwise instructions, documents templates if available, and highlights relevant legal references.

3.2.3 Functional Requirements

- **REQ-5:** The system shall display step-by-step procedural guidance.
- **REQ-6:** The system shall provide links to official forms or documents where applicable.
- **REQ-7:** The system shall alert users if certain procedures require a licensed legal professional.

3.3 Multilingual Support

3.3.1 Description and Priority

Allows users to interact with the chatbot in English, Hindi, and other supported regional languages.
Priority: Medium

3.3.2 Stimulus/Response Sequences

- **Stimulus:** User selects a preferred language or submits a query in that language.
- **Response:** System responds in the selected language with equivalent legal explanations.

3.3.3 Functional Requirements

- **REQ-8:** The system shall detect the input language automatically or allow manual selection.
- **REQ-9:** The system shall generate responses in the selected language while preserving legal accuracy.
- **REQ-10:** The system shall handle transliteration and common legal terms appropriately.

4. External Interface Requirements

4.1 User Interfaces

The AI Chatbot for Legal Advice will provide a responsive web-based dashboard with modules for Legal Queries, Procedural Guidance, Legal Document Templates, and FAQ.

- **Input:** Text and voice queries, language selection.
- **Output:** Legal explanations, statutory references, procedural guidance.
- **Buttons/Functions:** Submit, Clear, Help, Settings.
- **Error Handling:** User-friendly messages for invalid inputs or API errors.

4.2 Hardware Interfaces

The AI Chatbot operates on standard desktops, laptops, tablets, and smartphones with internet access.

- **Input:** Keyboard, touchscreen, microphone (for voice queries)
- **Output:** Display on screen
- **Communication:** HTTPS requests to back-end APIs; optional WebSockets for real-time updates
- **Requirements:** Modern browser with JavaScript support; stable internet connection

No specialized hardware is needed beyond typical consumer devices.

4.3 Software Interfaces

The AI Chatbot interacts with the following software components:

- **OpenAI APIs:** Receives user queries (text/voice), returns legal explanations and references via HTTPS.
- **Whisper API:** Converts voice input to text for query processing.
- **LangChain:** Orchestrates workflows, passes queries and responses between APIs and database.
- **Database (SQLite/PostgreSQL):** Stores user queries, session history, and legal references.
- **Browser/OS:** Hosts the web interface, sends input, and displays responses.

All communications use secure protocols, and shared data includes query text, session info, and generated responses.

4.4 Communications Interfaces

The AI Chatbot communicates securely using:

- **HTTPS:** For client-server interactions and API calls (OpenAI, Whisper).
- **Email:** Optional encrypted notifications via SMTP/TLS.
- **Push Notifications:** Secure browser or mobile alerts.
- **Data Format:** JSON for queries and responses.
- **Security:** TLS encryption ensures confidentiality and integrity.
- **Synchronization:** Real-time responses with asynchronous support for longer tasks.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- **Response Time:** Legal queries and document generation within 5 seconds.
- **Concurrency:** Support at least 50 concurrent users initially.
- **Scalability:** Maintain stable performance as user and query volume grows.
- **Error Handling:** Notify users clearly if system is busy or overloaded.

5.2 Safety Requirements

- **Data Protection:** User journals, planner entries, and queries shall be auto-saved and backed up regularly.
- **Data Loss Prevention:** No action shall cause permanent data loss without explicit user confirmation.
- **Access Control:** Only authorized users may access personal data; data must be encrypted in storage and transit.
- **Regulatory Compliance:** Design must follow relevant data privacy regulations (e.g., Indian IT Act, GDPR if applicable).

5.3 Security Requirements

- **Authentication:** Secure login via JWT tokens or session-based methods.
- **Authorization:** Users access only their own data; role-based access for admins.
- **Data Protection:** Encrypt sensitive data in storage and transit; maintain regular backups.
- **Session Management:** Sessions expire after inactivity; detect unauthorized access.
- **Compliance:** Follow Indian IT Act, GDPR (if applicable), and industry security best practices.

5.4 Software Quality Attributes

- **Usability:** Intuitive dashboard for non-technical users.
- **Scalability:** Supports growing user base and query volume.
- **Maintainability:** Modular design for easy updates.
- **Security & Privacy:** Data encrypted; access controlled.
- **Performance:** Queries and document generation ≤5 seconds.
- **Portability:** Works on Windows, macOS, Linux, and modern browsers.
- **Interoperability:** Seamless integration with APIs and databases.

5.5 Other Requirements

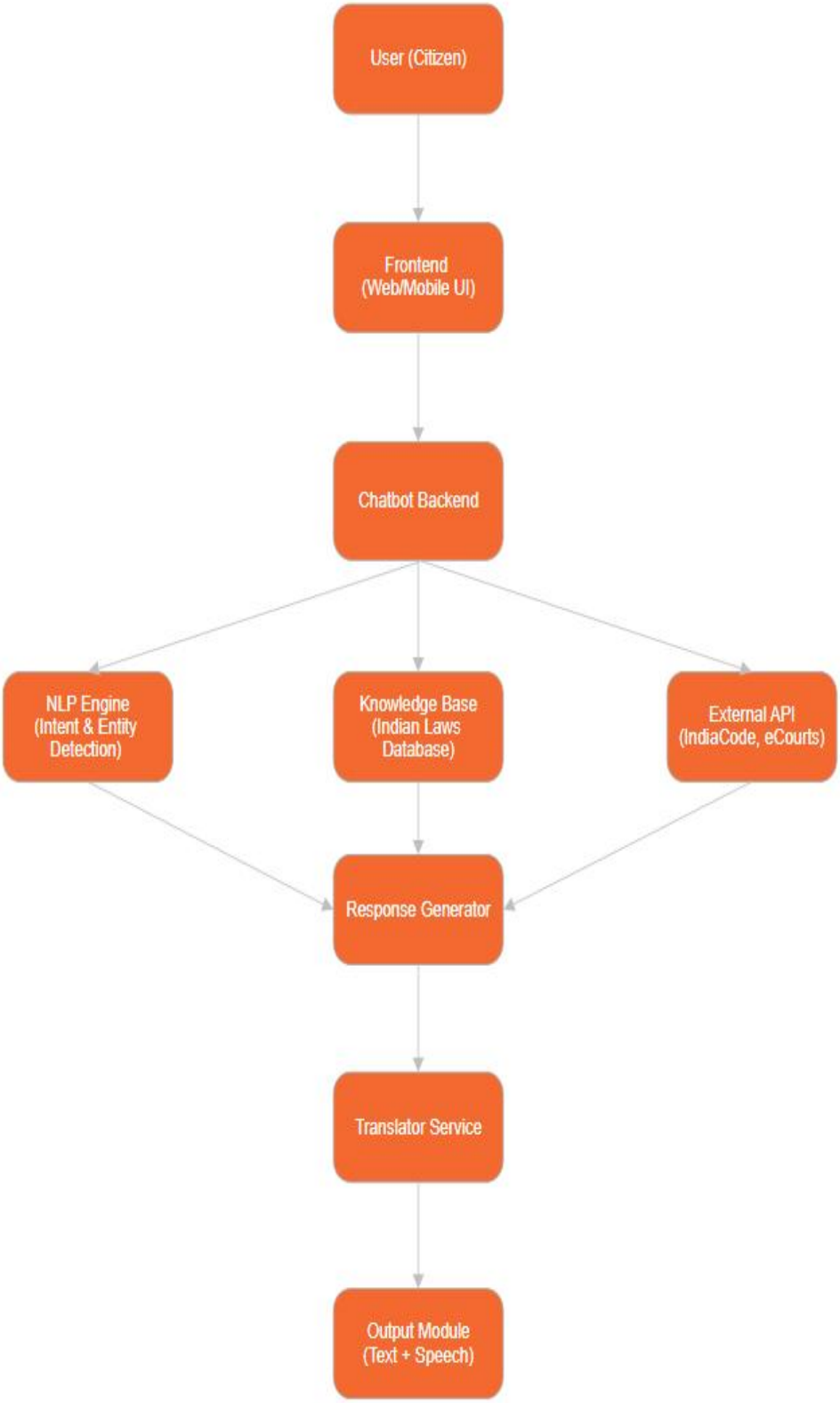
- **Data Privacy:** Comply with GDPR-like principles and Indian IT Act; encrypt all sensitive data.
- **Internationalization:** Support multiple languages (English, Hindi, Marathi) and future language additions.
- **Database:** Use SQLite/PostgreSQL with backups and transactional integrity.
- **Compliance:** Follow Indian laws and regulations for legal information and AI services.
- **Future Integration:** Architecture allows easy addition of tools, APIs, or cloud storage.

Appendix A: Glossary

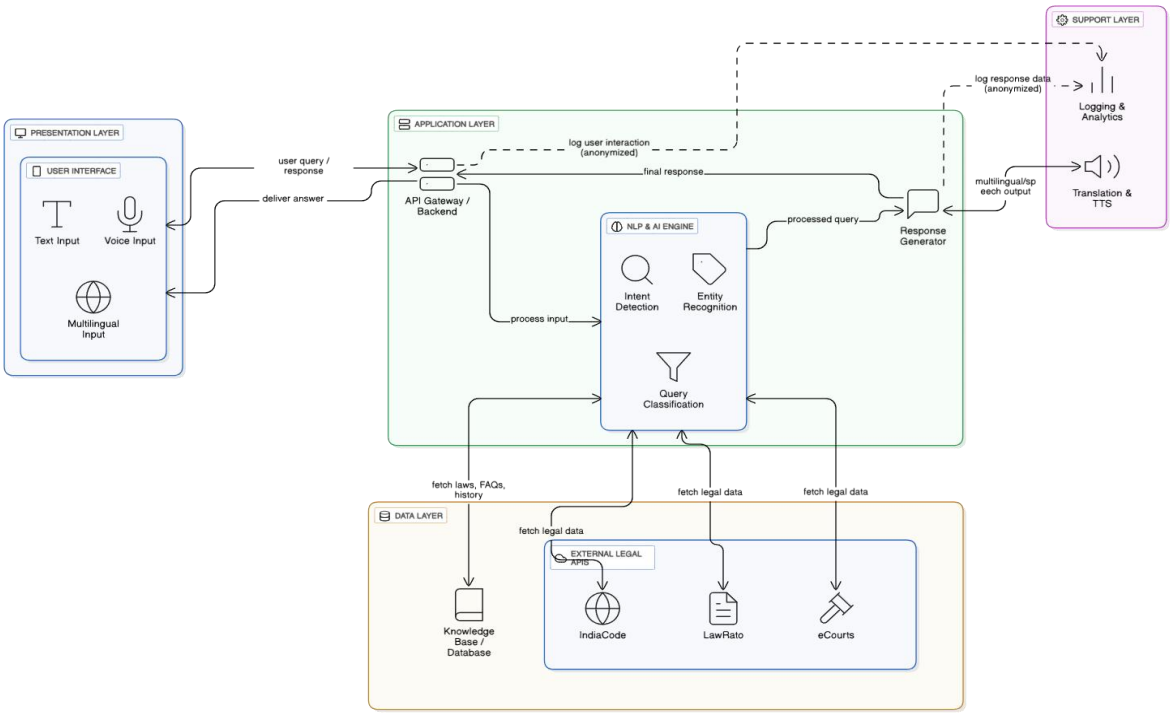
- **AI API:** External service providing AI-generated legal responses and insights.
- **Whisper API:** Automatic speech-to-text service for converting user voice queries to text.
- **JWT (JSON Web Token):** Standard for secure user authentication and session management.
- **LangChain:** Framework for orchestrating multi-step AI workflows and API interactions.
- **GDPR:** General Data Protection Regulation, governing data privacy standards.
- **IT Act (India):** Information Technology Act, 2000, regulating electronic data and privacy in India.
- **Legal Query:** A user-submitted question regarding laws, procedures, or statutory interpretation.
- **User Session:** The period in which a user interacts with the chatbot before logging out or timing out.

Appendix B: Analysis Models

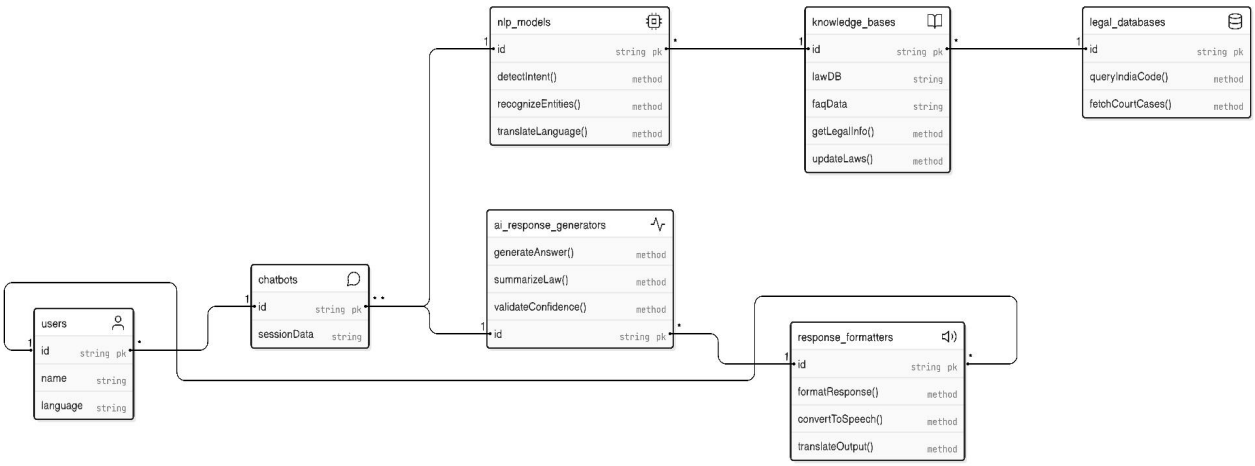
B.1) 1.System Design



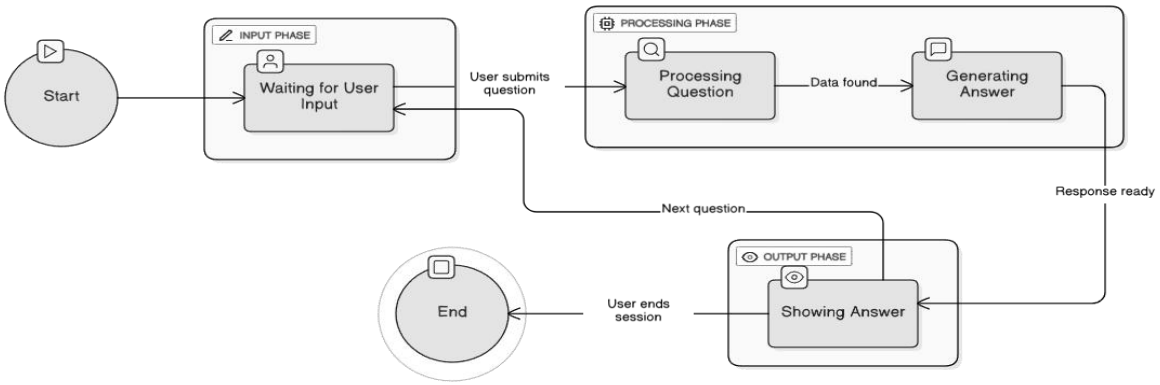
2. Component Architecture



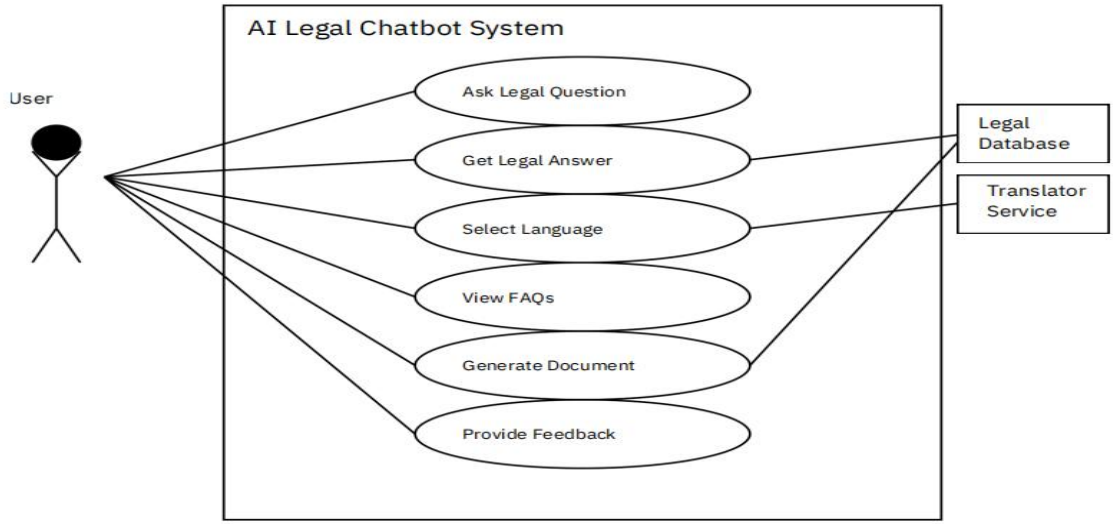
B.2) UML Class Diagram



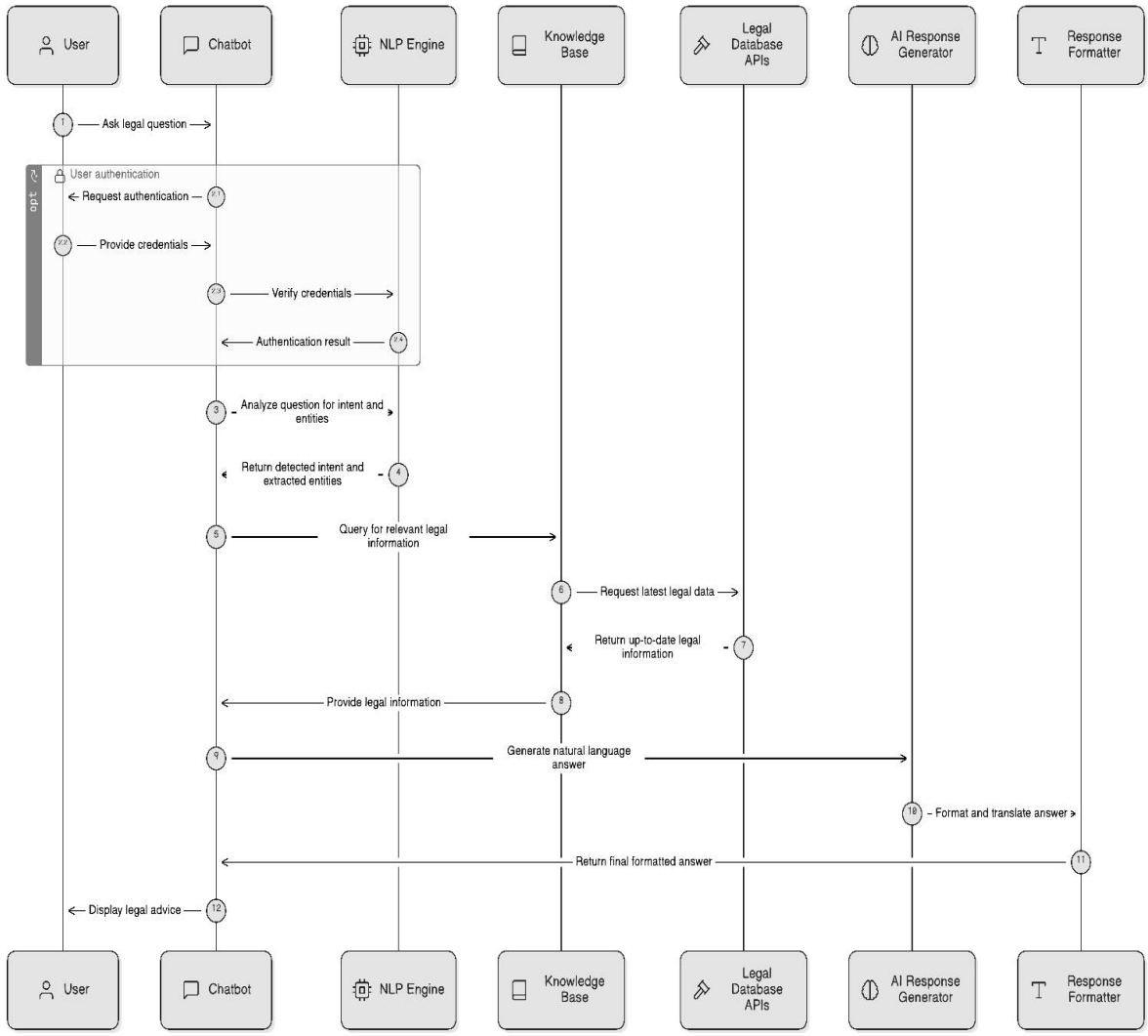
B.3) UML State Diagram



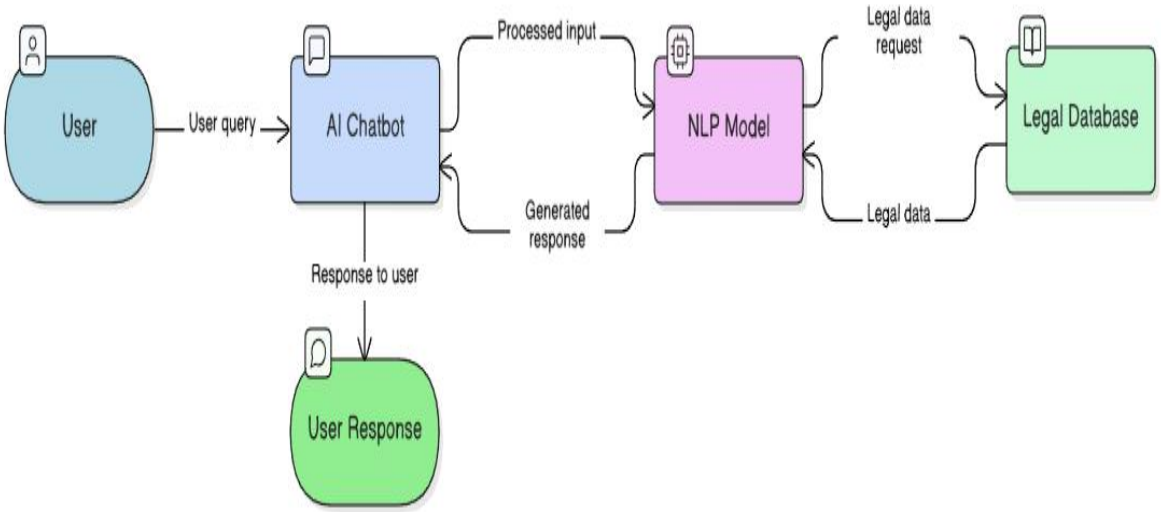
B.4) UML Case Diagram



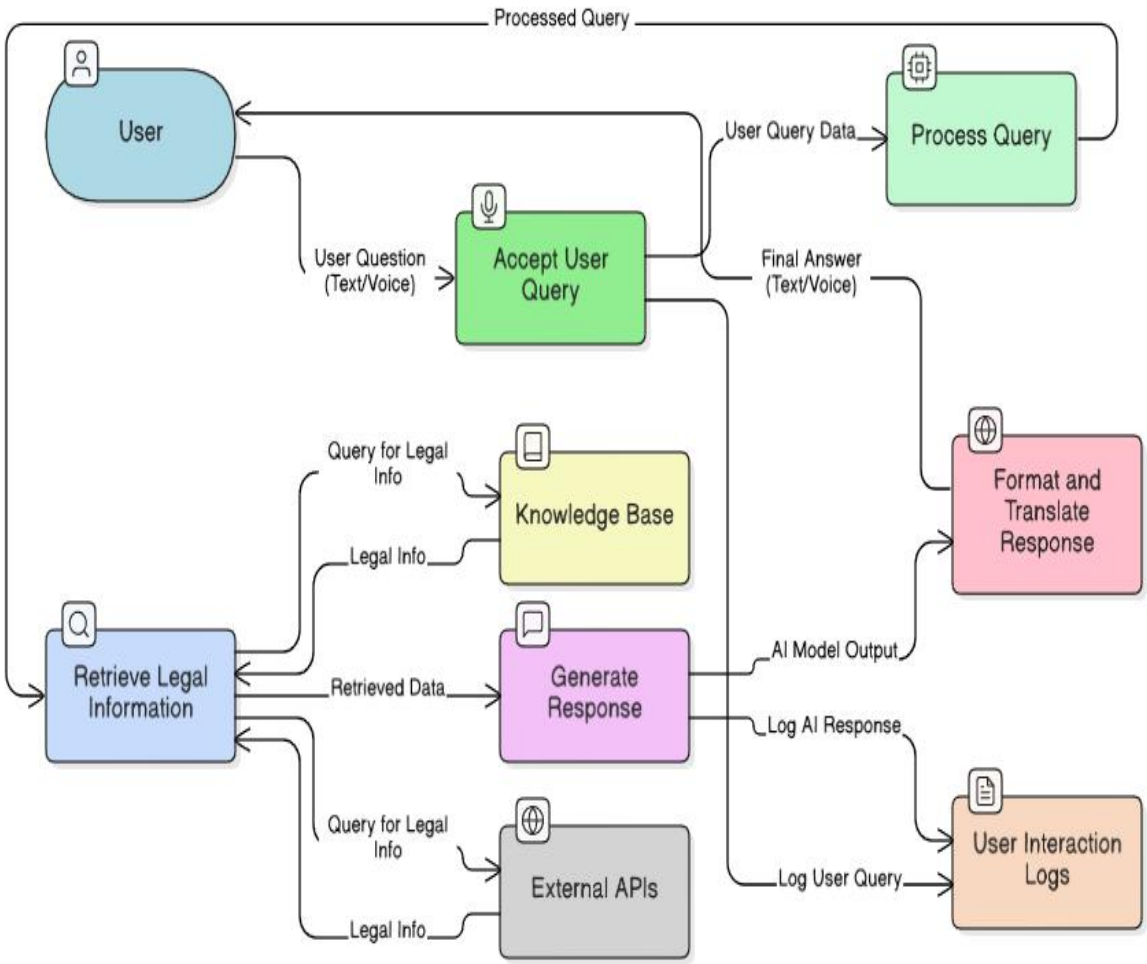
B.5) UML Sequence Diagram



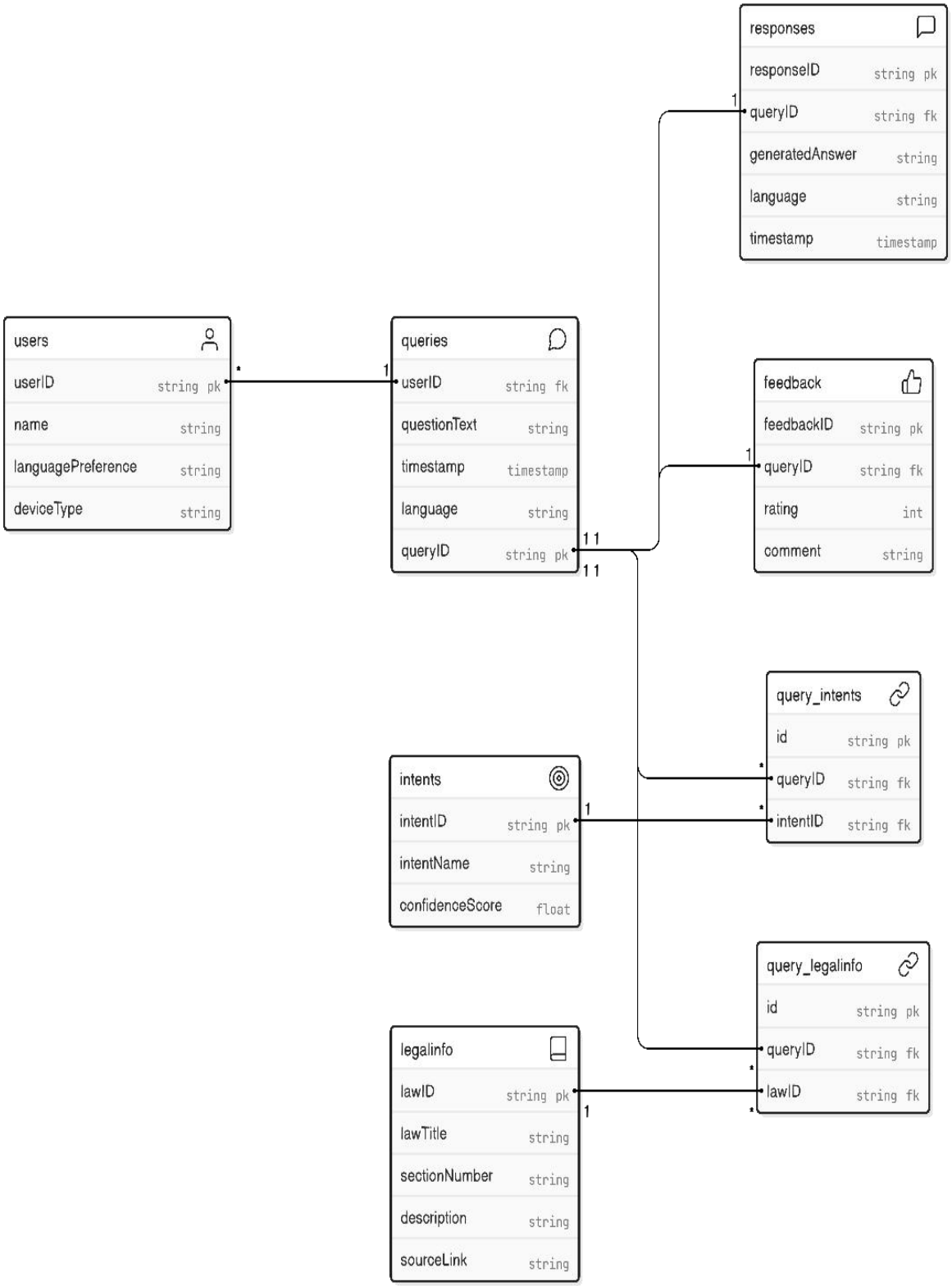
B.6) DFD Level 0



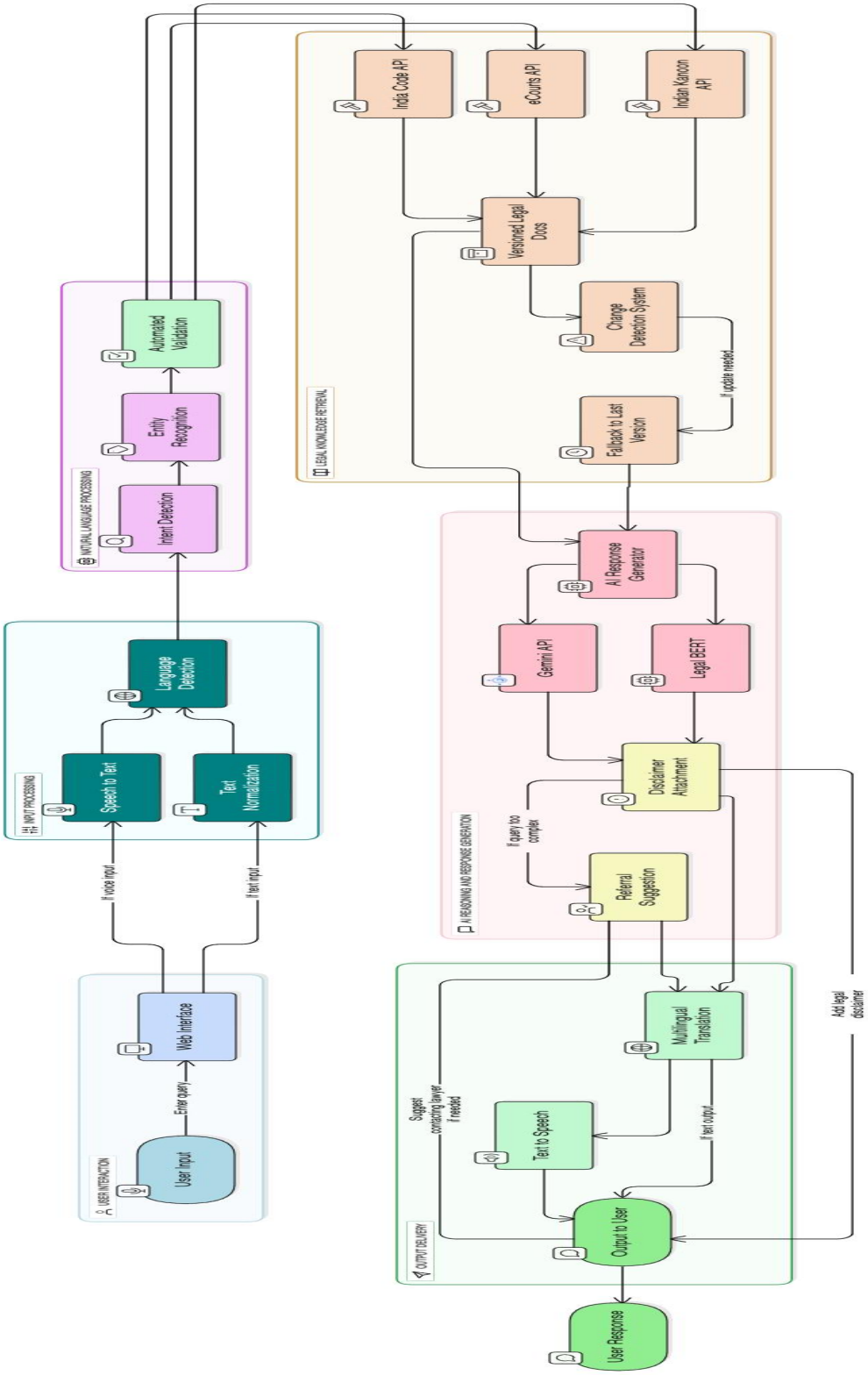
B.7) DFD Level 1



B.8) ER Diagram



B.9) DFD Level 2



Appendix C: Issues List

This list captures open issues and pending decisions for the *AI Chatbot for Legal Advice*:

- **Planned Features:** Advanced multilingual support beyond English, Hindi, and Marathi.
- **Integration Decisions:** Potential future integration with government legal databases or cloud storage.
- **Regulatory Clarifications:** Compliance requirements for AI-generated legal guidance and disclaimers.
- **Performance Targets:** Exact concurrency limits and scaling thresholds for large user bases.
- **UI/UX Details:** Finalization of accessibility enhancements and keyboard shortcuts.
- **Data Retention Policies:** Duration and management of user journals, queries, and session history.

  This list is dynamic and will be updated as issues are resolved or clarified during development.